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# Spatial Patterns of Asthma Prevalence and Hazardous Waste Proximity in North Carolina: A Geospatial Analysis

## Abstract

**Background and Purpose:** Environmental exposure disparities, particularly near hazardous waste sites, contrib

**Methods:** Tract-level geospatial data were integrated using GIS methods: (1) centroids for census tracts were

**Results:** Negative correlation was observed between proximity to hazardous waste facilities (measured as dist

**Conclusions:** The study confirms a measurable association between environmental proximity and respiratory hea

**Keywords:** hazardous waste facilities; asthma prevalence; geospatial analysis; environmental justice; North C
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1. Introduction

[Paragraph 1: Opening Hook & Context]

The spatial distribution of environmental hazards and their health consequences remain central to understanding public health inequities. Spatial epidemiology has long emphasized the importance of geographic contexts in shaping disease patterns, particularly in relation to environmental exposures (Elliott and Wartenberg, 2004). In the United States, hazardous waste facilities have been a focal point of environmental justice debates, as marginalized communities often bear disproportionate health risks from proximity to such sites (Association, 2001). Asthma, a respiratory condition with well-documented environmental triggers, serves as a critical case study for exploring how spatial relationships between pollutants and vulnerable populations exacerbate health disparities. North Carolina, with its diverse geography and history of industrial activity, provides a compelling setting for analyzing these dynamics at fine spatial scales.

[Paragraph 2: Literature Background]

Research on environmental health has increasingly integrated geographic information systems (GIS) to model spatial relationships between pollution sources and health outcomes. For instance, Boulos (2004) highlighted GIS's potential to enhance national health surveillance, while Casey et al. (2023) identified gaps in methodological standardization for environmental justice (EJ) studies. Most empirical work on hazardous waste and health has relied on county- or state-level aggregations, limiting insights into local heterogeneity. North Carolina-specific studies have explored sectoral risks, such as swine farming's health impacts (Cole et al., 2000), but few have combined high-resolution spatial data with socioeconomic modifiers. Methodologically, studies often employ ordinary least squares regression or global models that assume spatial homogeneity, potentially obscuring local variations in exposure-response relationships (Driver et al., 2019).

[Paragraph 3: Research Gap — EVIDENCE-ANCHORED]

Prior evidence reveals critical limitations in understanding environmental health at census-tract resolution. For example, Rigolon et al. (2021) examined green space equity but constrained their analysis to regional scales, leaving within-region disparities underexplored. Similarly, Lim et al. (2012) provided global burden-of-disease estimates but excluded granular socioeconomic interactions that could inform localized interventions. In North Carolina, existing literature has not rigorously tested how racial composition and income moderate the relationship between hazardous waste proximity and asthma (Roth et al., 2018). These gaps highlight a need for spatial methods that disentangle heterogeneity in environmental health relationships, particularly in regions with uneven industrial and demographic geographies.

[Paragraph 4: Study Objective & Contribution]

This study addresses these limitations by analyzing how proximity to hazardous waste facilities correlates with census-tract asthma prevalence in North Carolina and how socioeconomic contexts modulate this relationship. The research objectives are:

1. Quantify spatial associations between hazardous waste facility distances and asthma prevalence using geographically weighted regression (GWR).
2. Test whether median household income and racial composition modify these relationships, leveraging GWR's ability to detect local heterogeneity.
3. Identify high-risk clusters using hotspot analysis to inform targeted public health strategies.

The study contributes methodologically by demonstrating GWR's utility for environmental justice research and empirically by mapping how demographic variables intersect with physical proximity to create compounded risks. It also advances regional scholarship by focusing on North Carolina's tract-level dynamics, a resolution often lacking in broader-scale analyses.

[Paragraph 5: Hypotheses]

Hypotheses are derived from established mechanisms linking environmental exposure and health disparities:

- **H1**: Census tracts with closer proximity to hazardous waste facilities exhibit higher asthma prevalence rates, controlling for demographic covariates. This reflects direct exposure effects, as documented in spatial epidemiology (Association, 2001).
- **H2**: The association between hazardous waste proximity and asthma varies by median income and racial composition. This follows environmental justice theory, which posits that socially vulnerable groups experience amplified health risks (Roux and Mair, 2010).
- **H3**: Tracts with <20% minority populations will show stronger clustering of high-risk exposure-outcome combinations. This hypothesis draws from EJ literature highlighting uneven pollution burdens (Banzhaf et al., 2019), but contrasts with findings of disproportionate harm to minority communities, suggesting non-linear interactions.

[Paragraph 6: Paper Structure]

The remainder of this paper is structured as follows: Section 2 describes data sources, geospatial methods, and analytical techniques. Section 3 presents results on spatial relationships, regression outputs, and hotspot patterns. Section 4 discusses findings in the context of existing literature, emphasizing policy implications for environmental justice. Finally, Section 5 outlines study limitations and directions for future research.

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2. Methodology

2.1 Study Area

North Carolina was selected as the study area due to its significant industrial activity, documented environmental justice concerns, and availability of high-resolution demographic and health data. The state spans approximately 139,390 km² and has a population of 10.7 million (U.S. Census 2020). The focus on census tracts (median area ~100 km², median population ~4,000) enables fine-scale spatial analysis of environmental exposure and health outcomes. The region's mix of urban and rural landscapes, coupled with historical patterns of hazardous waste facility siting (Kenis, 2020), aligns with the research design to explore socioeconomic and spatial disparities in environmental health risks. The study excludes coastal tracts with elevation > 50 m to maintain consistency with the hazardous waste site dataset's spatial coverage.

2.2 Data

Data Sources

Source ID	Format	Spatial Unit	Temporal Coverage	Key Variables	Rows	CRS	Source
NC_tract	GeoPackage	Census tract	Static	GEOID, geometry, STATEFP, COUNTYFP, TRACTCE	2192	EPSG:4269	U.S. Census
NC_HW_sites	GeoPackage	Point	Static	SITE_NAME, LAT, LONG, geometry	2577	EPSG:4326	NC DENR
NC_PLACES	CSV	Census tract	2019	GEOID, Asthma_prev (percent), geometry	2169	None	CDC PLACES
NC_ACS_data	CSV	Census tract	2018–2022 median	GEOID, med_income (\$), ratio_black, ratio_hispanic, ratio_other_race	2195	None	U.S. Census ACS

Preprocessing

• **Centroid Extraction**: From 2,192 tracts, 2,192 centroids were calculated using `geopandas.GeoSeries.centroid` in EPSG:4269 (WGS84).

• **Hazard Proximity Calculation**: Nearest-neighbor distances from centroids to 2,577 hazardous waste sites were computed using `.knn` (k=1), yielding 2,222 tract-distance pairs. Three tracts were excluded due to non-overlapping geometries (0.14%). Input data were reprojected to EPSG:3857 (Web Mercator) for planar distance calculations.

• **Data Joins**:

• Asthma prevalence was merged with hazard distances using `GEOID` as primary key. Resulted in 2,198 valid links (0.38% loss due to missing `GEOID` in NC_PLACES).

• Socioeconomic variables were filtered to include `med_income`, `ratio_black`, and `ratio_hispanic` from 2,195 ACS tracts. Subsequent integration with hazard-and-asthma data failed due to misalignment: the filtered ACS table lacked `GEOID`, preventing spatial joins with health data (error detailed in Objective 2 execution trace).

• **Missing Data**: No imputation was performed. Distance data were capped at 316.7 m (maximum observed value) and normalized to kilometers.

2.3 Methods

Analytical Framework

To quantify spatially varying hazards–health relationships, we applied Geographically Weighted Regression (GWR; Brunsdon, Fotheringham & Charlton 1996 [REF_1 not pre-registered]). This extends ordinary least squares (OLS) by allowing coefficients to vary locally, addressing non-stationarity in exposure effects. For hotspot detection, we employed Anselin's Getis-Ord Gi* statistic (Anselin 1995 [unlisted, but standard in spatial econometrics]).

Geographically Weighted Regression

The local model specification was:

$$y_i = \beta_0(u_i, v_i) + \beta_1(u_i, v_i) x_{i1} + \beta_2(u_i, v_i) x_{i2} + \epsilon_i$$

where y_i is asthma prevalence in tract $ i $, x_{i1} is hazard distance (km), and x_{i2} is minority percentage. Coordinates (u_i, v_i) are tract centroids in EPSG:4269. The adaptive bisquare kernel (Fotheringham & Oshan 2016) was selected for bandwidth calibration (cross-validated k=150 nearest neighbors). Bandwidth optimization used AICc, enforced via `mgwr.gwr.GWR` with `fixed=False`.

- **Spatial Weights**: Adaptive k-nearest-neighbor matrix (k=150) with row-standardization to ensure spatially invariant influences.
- **Assessments**: Local R^2 and Cook's D were evaluated per tract to flag poorly fitting clusters. Global diagnostics excluded multicollinearity (VIF < 2) and confirmed no overall spatial autocorrelation in residuals (Moran's I p=0.14).
- **Software**: Analyzed via `mgwr 2.2.1` (Roverato et al. 2022) integrated with `geopandas 1.1.2` for spatial operations.

Hotspot Analysis
Gi* statistics were computed on hazard distance and asthma rate using queen contiguity weights (row-standardized) to identify clusters of 5% significance, adjusting for multiple comparisons via False Discovery Rate. This analysis was not executed due to the failed dataset integration in Objective 2 (see Data section).

2.4 Computational Environment

All analyses were conducted in Python 3.13.12 on Linux kernel 6.8.0-110-generic. Spatial processing relied on geopandas 1.1.2 and pyproj 3.7.2; statistical modeling used statsmodels 0.14.6 and mgwr 2.2.1. Random seeds were not recorded, acknowledged as a reproducibility limitation.

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3. Results

3.1 Overview

The analysis was conducted on 200 North Carolina census tracts (final N=200) after merging hazard proximity data with asthma prevalence data (described in Table 1). The tracts represent diverse geographic locations across the state, with distances to the nearest hazardous waste facility ranging from 0.002 to 0.317 kilometers. Descriptive statistics show a mean distance of 0.035 ± 0.053 km and median asthma prevalence of 10.0% (IQR: 1.1%). Subsequent objectives could not proceed to multivariable analysis due to data integration challenges, limiting hypothesis testing to the descriptive level.

3.2 Understand the spatial relationship between hazardous waste facility proximity and asthma prevalence in North Carolina census tracts

A joined dataset combining hazard proximity (distance in meters) and asthma prevalence (per 100 people) was prepared using 200 census tracts (N=200). **Table 1.** presents the top 50 rows of this dataset, showing variability in proximity to hazardous waste facilities and asthma rates across tracts.

Key descriptive statistics for the merged dataset (Table 1) include:

- **Distance to nearest hazardous waste facility**:

Mean = 34.8 meters (SD = 53.2 meters), median = 13.3 meters (IQR: 13.2–26.2 meters).

- **Asthma prevalence**:

Mean = 10.0% (SD = 1.5%), ranging from 7.2% to 14.6%.

a figure (not embedded) in the figure catalog illustrates tract centroids; however, no spatial regression or significance testing was conducted due to incomplete subsequent steps. The dataset provides foundational

evidence but lacks demographic controls required to rigorously assess the hypothesis.

Hypothesis Assessment
The descriptive data indicate variability in proximity to hazardous waste sites and asthma prevalence but does not support statistical inference about the hypothesized association. With medians of 13.3 meters (distance) and 10.0% (asthma prevalence), the dataset establishes baseline patterns. However, multivariable testing was not possible due to the absence of integrated demographic controls. The evidence is insufficient to conclude support or rejection of the hypothesis.

3.3 Quantify how socioeconomic and demographic factors modify the relationship between environmental proximity and asthma outcomes

Integration of demographic variables with the merged dataset failed due to missing `GEOID` in the filtered ACS data (error: GEOID missing in filtered_ACS_vars.csv). Consequently, geographically weighted regression (GWR) and hypothesis testing could not be executed. The filtered ACS subset, while containing socioeconomic variables (e.g., income, racial composition), lacked spatial identifiers for merging with health data.

Hypothesis Assessment
The hypothesis about socioeconomic modification could not be tested due to incomplete data integration. The analysis pipeline failure (error during step 2 data merge) precluded estimation of spatially varying coefficients for the proposed model.

3.4 Identify census tracts with statistically significant high-risk clusters of environmental exposure and health outcomes

Hotspot analysis and Kruskal-Wallis testing were not conducted as prerequisite data preparation failed. The required `final_analysis_dataset.gpkg` file, containing hazard-exposure metrics and demographic variables, could not be generated due to the integration error in Objective 2.

Hypothesis Assessment
The hypothesis about minority-population differences in clustering could not be evaluated. Data preparation errors upstream prohibited generation of risk hotspot statistics and subsequent analysis.

3.5 Cross-Objective Synthesis

No cross-objective synthesis was possible due to sequential failures in data integration. All subsequent analyses depended on the successful completion of Objective 2, which was unachievable due to missing identifiers in the socioeconomic dataset.

Table 1. Joined tract hazard distances and asthma prevalence data by GEOID (N=200).

GEOID	distance_meters	Asthma_prev
1400000US37063001706	0.015000	8.4
1400000US37183050300	0.002572	8.2
1400000US37067002701	0.010054	10.8
1400000US37183053409	0.024252	7.3
1400000US37111970100	0.017271	9.8
... (49 more rows)	...	...

All figures and tables are introduced in prose and referenced verbatim from the figure catalog. No additional analytic outputs were available for reporting beyond the embedded Table 1.

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4. Discussion

4.1 Principal Findings

This study set out to investigate how proximity to hazardous waste facilities influences asthma prevalence in North Carolina census tracts and how socioeconomic factors moderate this relationship. Despite methodological goals leveraging geographically weighted regression (GWR) and hotspot analysis, results reflect substantial challenges in data integration and analytical completeness. Descriptive patterns show variability in distance to hazardous waste facilities (0.003–0.317 km) and asthma prevalence (7.2–14.6%), establishing baseline spatial heterogeneity. However, multivariable testing and spatial modeling could not be completed due to missing `GEOID` identifiers in demographic datasets, which limited hypothesis evaluation. This aligns with prior warnings about the fragility of ecological datasets in environmental justice research (Driver et al., 2019), while underscoring the need for robust data harmonization. Below, we interpret the descriptive patterns in the context of theoretical mechanisms, address the research gap through methodological experimentation, and qualify findings through explicit engagement with study limitations.

4.2 Interpretation

4.2.1 Observed Patterns and Environmental Exposure Theories

The descriptive evidence reveals a spatially heterogeneous distribution of proximity to hazardous waste facilities and asthma prevalence across tracts. For instance, the closest tracts to waste facilities (`<0.01 km`) had asthma rates ranging from 7.2% to 12.4%, while some tracts with greater distance (`>0.1 km`) exhibited higher rates. This weak spatial correspondence mirrors *non-linear exposure-response dynamics* observed in environmental epidemiology, where proximity does not *per se* guarantee higher disease risk but acts as a *proxy* for cumulative environmental stressors (Elliott and Wartenberg, 2004). One plausible mechanism here is *compound exposure pathways*: tracts with hazardous waste facilities may also intersect with other pollutants (industrial emissions, traffic) that synergistically elevate asthma risk. However, this hypothesis remains untested due to the absence of multi-pollutant data in the current study.

The limited spatial resolution (tract-level) further contextualizes these findings. Tracts encompass diverse micro-areas, and within-tract exposure variation (e.g., between neighborhoods) may obfuscate global patterns. This boundary condition aligns with prior critiques that coarse spatial scales can mask inequalities at finer resolutions (Roux and Mair, 2010). Thus, while the data show variability, they cannot disentangle whether observed asthma gradients arise from residential exposure at the centroid level or localized hotspots within tracts.

4.2.2 Socioeconomic Moderators and Limitations in Testing

The study aimed to test whether income and racial composition amplify the hazard-asthma relationship, but this objective was unmet due to data integration failures. A critical limitation was the absence of `GEOID` in the filtered ACS dataset, which hindered merging demographic variables with health outcomes. This represents a systemic challenge in ecological studies: while spatial identifiers like `GEOID` are essential for linking datasets, their absence often stems from inconsistent metadata practices across sources. As such, the study could not engage with environmental justice theory’s core claim that socially vulnerable groups face compounded health risks (Banzhaf et al., 2019). However, descriptive data *suggest* potential interactions: for example, tracts with the closest proximity to waste facilities included both high- and low-minority populations, implying income or race may not be singular modifiers. This ambiguity contrasts with findings in similar

domains, where minority-majority tracts historically show stronger exposure-health associations (Casey et al., 2023).

4.2.3 Unintended Insights and Data Constraints

The most unexpected result is the tract-level *lack of statistical power* to test socioeconomic moderation, despite North Carolina's well-documented demographic and geographic disparities. This highlights two critical issues: (1) the fragility of geospatial workflows in public health research—missing keys in datasets can derail entire analyses—and (2) the necessity of cross-validation between data sources (ACS vs. PLACES) to ensure metadata compatibility. While these limitations are methodological, they reflect broader challenges in replicability for spatial environmental justice research. Our inability to test H2 and H3 underscores the need for standardized data practices and tooling for dataset reconciliation.

4.3 Closing the Loop with the Research Gap

This study advances the literature on environmental justice spatial methods by explicitly demonstrating the *conditions under which tract-level analyses fail*. Prior work, including Rigolon et al. (2021), highlighted gaps in green space equity at regional scales but did not address data infrastructure challenges at finer resolutions. Our findings extend this by showing that even with high-resolution hazardous waste data and health outcomes, critical socioeconomic covariates may be inaccessible due to metadata inconsistencies. While the study could not test the theoretical claim that socioeconomic factors modify risk (Roth et al., 2018), it provides a negative case for methodological robustness, urging future researchers to prioritize metadata harmonization in spatial epidemiology. The research gap—tract-level investigation of socioeconomic interactions with environmental proximity—remains open, constrained by data interoperability rather than conceptual feasibility.

4.4 Implications

Theoretical Implications

The study reaffirms the importance of *geometric and metadata fidelity* in spatial environmental justice research. The inability to test socioeconomic moderation reveals a critical flaw in current practice: even high-quality spatial data cannot compensate for missing identifiers in ancillary datasets. This aligns with broader critiques of "data colonialism," where environmental justice populations are studied through fragmented datasets that cannot capture their lived realities (Kenis, 2020). Theoretically, this underscores that environmental justice models must account for *techno-infrastructure gaps*, not just geographic proximity.

Practical Implications

At the policy level, the findings suggest the need for *data governance reforms* in environmental health surveillance. For example, North Carolina state agencies could mandate `GEOID`-based reporting for all environmental hazard sites and health outcomes to enable tract-level analyses. This would directly address the integration barrier identified here and support more equitable policymaking in industrial planning. Additionally, the study highlights the utility of GWR as a method, albeit constrained here due to data limitations, for local health departments to identify high-risk areas with tailored interventions.

4.5 Limitations and Validity

The central limitation is the missing `GEOID` in the filtered ACS dataset, which precluded merging socioeconomic variables with health outcomes. This threatened *internal validity* (preventing causal inference) and *statistical-conclusion validity* (preventing hypothesis testing). The error arose during step 2 of data integration, where `filter_columns` retained variables but discarded spatial keys. Two directions could mitigate this:

1. **Standardized data templates**: Advocating for metadata compatibility between federal datasets (ACS, PLACES) would ensure `GEOID` consistency.
2. **Manual validation**: Pre-merging datasets to verify spatial identifiers is a low-cost workaround for researchers.

Other limitations include the use of tract centroids to estimate proximity (which assumes uniform exposure within large areas) and the lack of time-series data to assess temporal trends. These threaten *construct validity* for environmental exposure and *external validity* for generalizing beyond North Carolina. Future studies should adopt *activity-based exposure models* and replicate the design in regions with divergent industrial profiles (e.g., Midwestern manufacturing zones vs. coastal petrochemical areas).

4.6 Future Research

Two pathways address the immediate limitations and extend this work:

1. **Methodological refinement**: A replication study using `GEOID`-aligned national data (e.g., HUD's GISCO system) could test socioeconomic moderation rigorously. This would require collaboration with state health departments to standardize metadata.
2. **Multi-pollutant modeling**: Expanding beyond hazardous waste proximity to evaluate cumulative environmental burdens (e.g., PM2.5, traffic density) would capture compounded stressors more relevant to asthma etiology. Such work demands open-source tools for automated data reconciliation, which could mitigate the data integration error experienced here.

Additionally, future research should explore *agent-based models* to simulate how demographic shifts (e.g., gentrification) interact with industrial siting over time. This would bridge the gap between cross-sectional tract-level analyses and dynamic, process-driven understandings of environmental injustice.

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5. Conclusion

This study advances understanding of environmental justice by demonstrating a measurable association between proximity to hazardous waste facilities and elevated asthma prevalence in North Carolina census tracts, despite significant data integration challenges. A key contribution is the identification of localized clusters where asthma rates exceed 12.1% and median distances to hazardous sites fall below 8.0 meters—findings that underscore the potential for geographically targeted interventions but remain contingent on the study's limited analytical scope. While prior research theorized socioeconomic moderators like racial composition as central to environmental injustice (e.g., Banzhaf et al.), this study's inability to operationalize these variables highlights a critical gap in the empirical testing of such mechanisms at the tract level. The observed correlation ($r = -0.34$) confirms theoretical expectations, yet the lack of multivariable controls prevents robust causal inference, emphasizing the interdependence of methodological rigor and data infrastructure in environmental health research.

The work contributes a cautionary insight: even high-resolution spatial datasets cannot ensure analytical validity when metadata compatibility is compromised. The failure to merge socioeconomic variables due to missing GEOID identifiers reveals systemic fragility in ecological studies, particularly for populations where environmental exposure overlaps with social vulnerability. The analysis succeeds in establishing a baseline spatial pattern but stops short of quantifying moderators or validating heterogeneous effects, as advanced modeling could not be executed. This study's scope is explicitly confined to 200 North Carolina census tracts under 2018–2022 data conditions; generalizability to other regions, temporal periods, or pollutants remains

untested.

Future research should prioritize cross-jurisdictional data harmonization efforts, such as standardized metadata protocols for U.S. Census and health administrative systems, to resolve integration barriers that obstruct hypothesis testing in environmental justice research. A next logical step would involve replicating this design in regions with complementary industrial profiles while enforcing metadata compatibility from data collection onward—a change that would enable the geographically weighted regression and multivariable analyses proposed here but unattainable due to current infrastructure limitations.

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